

# EGY 501 Course Outline

## EGY 501 — Energy Sector Modelling

### Course Outline

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|----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>School</b>        | 05 Sector Planning — Energy                                                                                                                                                                                    |
| <b>Level</b>         | 500 (practitioner / advanced)                                                                                                                                                                                  |
| <b>Hours</b>         | 55 (indicative band 40–55)                                                                                                                                                                                     |
| <b>Shape</b>         | 6 modules · 25 sessions · 12 laboratories · 4 assignments · 1 final assessment                                                                                                                                 |
| <b>Delivery</b>      | Instructor-led — online, blended or in person                                                                                                                                                                  |
| <b>Prerequisites</b> | EDIS 101 required. Linear programming at the level of reading a formulation; Julia at the level of reading and modifying a script; Python with pandas; survey weights at the level of DAT 703. No PyPSA needed |
| <b>Assessment</b>    | Pass 60 %, with 60 % laboratory competency                                                                                                                                                                     |
| <b>Version</b>       | 1.1 · reviewed 2026-09-04 · 12-month cycle                                                                                                                                                                     |

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### What this course is

EGY 501 teaches an energy-sector model estate and the judgement required to say what each of its models does and does not establish. The estate is 29 Julia modules across 13 layers, the power system solved in PyPSA: a system engine from demand through dispatch to the energy balance, nine non-power programmes, a GIS layer and six integration bridges, implementing a 24-part, 97-section specification.

Every number here was produced by running that estate on a real national case. Eight of its 28 registered limitations were found by re-deriving what it reported — a macro export discarding 81.2 % of the country's generation, a renewable share out by thirteen, a price out by 730 — and each is where the course spends its time.

**The rule the course is built on.** Every number has a model behind it and a denominator under it. Where the data cannot support the model, this course says so and teaches the specification anyway — knowing what you would need is part of knowing what you have.

### Who it is for

Ministry of Energy planners and power-system engineers; energy economists; electricity regulator staff; Ministry of Finance and statistical office analysts; and modellers joining an energy-sector planning team.

## Data and access

No learner touches restricted microdata: no laboratory opens a microdata file, and no identifier and no survey cluster coordinate appears anywhere. Two labels carry through every figure — estimated for a quantity computed from microdata, with its interval and its denominator, and specified for a model the catalogue defines but the data cannot support. Presenting the second as the first is an automatic fail. Restricted spatial layers are never redistributed: what ships is the reduction.

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## Structure

| Module    | Title                                                             | Sessions | Laboratory | Hours       |
|-----------|-------------------------------------------------------------------|----------|------------|-------------|
| <b>M1</b> | Foundations: the system, the specification, the data, the adapter | 01–05    | 01–02      | 10.5        |
| <b>M2</b> | The operating problem: demand, resource, dispatch, the balance    | 06–08    | 03–04      | 9.5         |
| <b>M3</b> | The physical system: networks, water, reliability                 | 09–11    | 05         | 8.0         |
| <b>M4</b> | People and money: access, households, tariffs, emissions          | 12–15    | 06–07      | 11.0        |
| <b>M5</b> | Beyond the sector: the CGE chain, poverty, validation             | 16–19    | 08–09      | 9.0         |
| <b>M6</b> | Method: gaps, scenarios, briefs, portability                      | 20–25    | 10–12      | 7.0         |
|           | Total                                                             | 25       | 12         | <b>55.0</b> |

*The 25 sessions and 12 laboratories, in teaching order, and the subject matter of each:*

**M1 Foundations: the system, the specification, the data, the adapter** — Sessions 01–05 · 10.5 h · Laboratory 01–02

- **01 What this system models** — model-estate architecture; the layers and what each produces; the boundary between a power-system model and a sector model
- **02 The specification, and what it specifies** — reading a technical specification; parts, sections and equations; implementation status; conformance of code to specification
- **03 The input schema, and what a column is worth** — tables, keys and units; required and optional inputs; provenance classes; the effect of a missing column
- **04 Geography, licence and a column that lied** — energy GIS layers: settlements, lines, plant registers, boundaries; data licensing and redistribution; intake checks

- **05 The adapter, and where units go to die** — from tables to a linear programme: sets, parameters, variables and constraints; time-slice weighting; units and conversion
- *Lab 01 The case as a set of claims* — case loading, intake diagnostics, and provenance classification of every column
- *Lab 02 Solve, and read the result in order* — running an economic dispatch; solver status, objective, primal and dual quantities, and the order in which they are read

**M2 The operating problem: demand, resource, dispatch, the balance** — Sessions 06–08 · 9.5 h · Laboratory 03–04

- **06 Demand: twelve periods stand for a year** — electricity demand and load profiles; load duration; representative time slices and weights; temporal aggregation; demand drivers
- **07 Resource, candidates and the boundary of the answer** — renewable resource assessment; capacity factors and availability; the candidate set; capital and operating costs; capacity credit
- **08 The number that did not add up** — the national energy balance; generation, losses and unserved energy; capacity against energy adequacy; energy-limited plant
- *Lab 03 Demand, resource and the boundary of the answer* — demand reconstruction from a load profile, and the sensitivity of a plan to the candidate set
- *Lab 04 The balance that closes to zero* — energy-balance accounting from input tables; graph connectivity; residual analysis

**M3 The physical system: networks, water, reliability** — Sessions 09–11 · 8.0 h · Laboratory 05

- **09 Networks: what clustering costs** — transport network representation: buses, lines, capacity, losses; spatial clustering; transmission expansion; congestion rent
- **10 Storage, water and the scenario that dominates** — storage state of charge, efficiency and duration; reservoir and run-of-river hydro; inflow years; climate stress testing
- **11 Reliability, trade and what cannot be validated** — reliability metrics (SAIDI, SAIFI, LOLE, value of lost load); cross-border trade and interconnection; their data requirements
- *Lab 05 Networks, water and the hydrological cases* — transmission expansion, hydrological scenario runs, and identifying decisions robust across them

**M4 People and money: access, households, tariffs, emissions** — Sessions 12–15 · 11.0 h · Laboratory 06–07

- **12 Access: a plan with one technology in it** — electrification planning: grid, mini-grid and standalone; least-cost geospatial allocation; budgets against access targets
- **13 Households: burden, ladder and the unconnected** — household energy survey analysis; the energy ladder and fuel stacking; energy poverty and burden; weights and denominators
- **14 Tariffs, finance and a test that cannot fail** — the utility revenue requirement: asset base, allowed return, depreciation; cost-reflective tariffs; lifeline cross-subsidy
- **15 Emissions and fuels: two accounts, two surprises** — power-sector greenhouse-gas inventory and emission factors; the household fuel account; carbon pricing and abatement response
- *Lab 06 Access: turn a cliff into a curve* — electrification modelling under a budget constraint and under an access target
- *Lab 07 Households, tariffs and who pays for the lifeline* — survey-weighted estimation of connection and burden; lifeline tariff design and subsidy incidence

**M5 Beyond the sector: the CGE chain, poverty, validation** — Sessions 16–19 · 9.0 h · Laboratory 08–09

- **16 The handoff, and where it fails** — energy-to-macro linkage; the GTAP-Power technology split; carrier mapping; price and productivity levers; interface contracts
- **17 From the CGE to three thousand households** — microsimulation from macro results; price and income transmission channels; FGT poverty measures; poverty lines and deflation
- **18 Validation: four right answers to one question** — cross-source reconciliation; source independence; a validation check against an identity; tolerances and reconciliation notes
- **19 Distribution: the denominator decides the story** — distributional and incidence analysis; the accounting unit; per-capita against per-household statistics; quintile reporting
- *Lab 08 Break and repair the CGE contract* — building and verifying a cross-model payload, and contract testing at an estate boundary
- *Lab 09 From the CGE to three thousand households* — microsimulation validation against a survey baseline, and FGT computation on the correct accounting unit

**M6 Method: gaps, scenarios, briefs, portability** — Sessions 20–25 · 7.0 h · Laboratory 10–12

- **20 The gaps register: using somebody else’s honesty** — limitations registers; classifying an entry by severity and reproducibility; model documentation and known-issue tracking
- **21 Designing a scenario set for a hydro system** — scenario design: axes, ranges and naming; structural against parametric uncertainty; sensitivity analysis; robust decisions
- **22 Writing the brief: six questions first** — communicating results to decision-makers; a recommendation as a decision rule; uncertainty and caveats; the structure of a brief
- **23 The fiscal chain: specified, not executed** — energy-sector fiscal linkages: subsidies, arrears, contingent liabilities, the MTEF budget bridge; reporting an absent link
- **24 Another country: what travels and what does not** — model portability; assessing data availability against a regional register; writing a data request with custodian and decision
- **25 Synthesis: the whole method, once more** — the full chain end to end, and the capstone brief
- *Lab 10 A register entry and a scenario set* — writing a limitations-register entry, and designing the scenario set that follows from it
- *Lab 11 Payloads, briefs and four sentences* — payload review, and writing a brief for a named recipient
- *Lab 12 Another country, and the capstone* — country data assessment against the register, and the capstone

Version 1.1 adds two further lecture sessions — 26 The Utopia case rebuilt on primary data, and 27 Utopia 2023: scenarios and money — which re-run the whole estate on a second case built from national returns. They sit outside the 55 audited hours and are taught as an extended case study or set as directed reading.

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## What a graduate can do

1. Load a national energy case, classify every value by provenance, and name the model that silently did not run for each absent optional table.
2. Solve an economic dispatch and apply the six checks in order, reporting passes as well as failures.
3. Re-derive a headline result from the input tables independently of the function that produced it, and close the energy balance.
4. Distinguish capacity adequacy from energy adequacy, and falsify a published explanation with the evidence that settles it.
5. Read a network result — including one that builds nothing — and say what it establishes.

6. Run a hydrological stress test, design a scenario set for a system whose dominant axis is structural, and recognise a corner solution.
7. Produce a survey-weighted estimate with its unweighted denominator and its disclosure decision.
8. Build a revenue requirement, design a cost-reflective tariff with a lifeline block, name who pays for it, and identify a validation check that is an identity.
9. Push a result across an estate boundary, verify the far side, and write its five contract tests.
10. Explain a null result rather than reporting it, write a register entry a stranger can reproduce, and close a brief with what it is, what it rests on, what it is not and what would change it.

*The sixteen formal learning outcomes, and the map from each to the assessment that tests it, are in 12\_LMS/learning\_outcomes.md.*

## Assessment

| Component                   | Weight | What it is                                                                                                                                                                                            |
|-----------------------------|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Knowledge quizzes</b>    | 15 %   | 139 items across 18 modules and seven question types                                                                                                                                                  |
| <b>Laboratory exercises</b> | 30 %   | 12 laboratories, equally weighted, marked on model competence, interpretation, honesty about limits and communication                                                                                 |
| <b>Applied assignments</b>  | 25 %   | The case-intake note; a solve read in order with one number re-derived; an incidence table on two denominators; a handover across an estate boundary                                                  |
| <b>Final assessment</b>     | 30 %   | Three hours, open book: definitions and vocabulary; calculation checkable against the shipped data; judgement about results this course found to be wrong; and a two-page brief for a named recipient |

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance sidecar is in the output: an irreproducible result is no result.

## Materials and set-up

A 261-page Technical Manual and a PyPSA model manual; 27 lecture decks and 12 laboratory decks; a Laboratory Manual, Instructor Guide and Learner Workbook; a 139-item quiz bank, four assignments and a final assessment with its marking guide; 93 datasets; 12 code exercises; 3 case studies; and the LMS package. The README lists every file.

Before the first laboratory, learners need the estate instantiated at `D:/EDIP/models/EnergyModelling`. On this host an Application-Control policy blocks native package images, so every Julia command in

this course passes `--pkgimages=no`. The full sequence is in the course README and in the Laboratory Manual's pre-flight section.

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*EGY 501 Energy Sector Modelling · version 1.1 · AICTEM EDIS Academy, School 05 Sector Planning.*